

Inexact Algorithms for Bilevel Learning

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August 10, 2023

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Variational Regularisation

Variational Regularisation

Variational regularisation problems have the form

$$\min_x \mathcal{D}(Ax, y) + R_\theta(x),$$

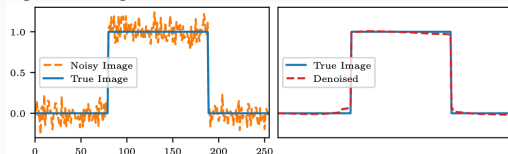
where we wish to find x given data $y \approx Ax$

Example (denoising): Given a noisy image y , find a denoised image x by solving:

$$\min_x \underbrace{\frac{1}{2} \|x - y\|^2}_{\mathcal{D}(x,y)} + e^{\theta_1} \underbrace{\sum_j \sqrt{\|\nabla x_j\|^2 + (e^{\theta_2})^2}}_{\text{smoothed TV}(x)} + \frac{e^{\theta_3}}{2} \|x\|^2$$

Solution depends on the choice of $\theta = [\theta_1, \theta_2, \theta_3]$:

$$\theta_1 = 0, \theta_2 = -6.9, \theta_3 = -6.9$$



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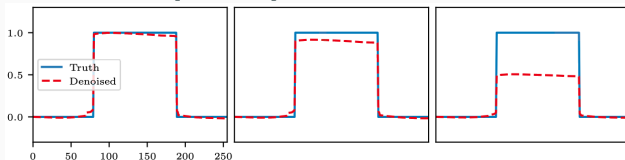
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Solution depends on the choice of $\theta = [\theta_1, \theta_2, \theta_3]$:

Vary θ_3

$\theta_1 = 0, \theta_2 = -6.9$



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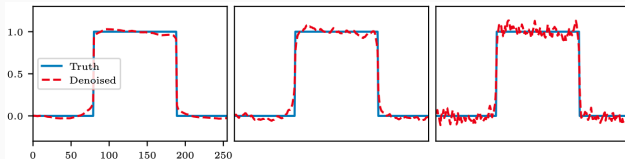
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Solution depends on the choice of $\theta = [\theta_1, \theta_2, \theta_3]$:

Vary θ_2

$\theta_1 = 0, \theta_3 = -6.9$



Variational Regularisation

Variational regularisation problems have the form

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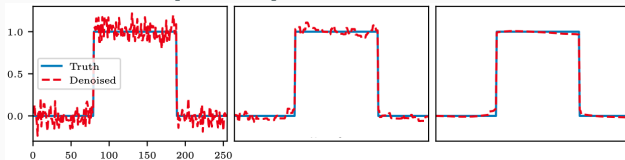
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Solution depends on the choice of $\theta = [\theta_1, \theta_2, \theta_3]$:

Vary θ_1

$$\theta_1 = -6.9, \theta_2 = -6.9$$



Choosing Parameters

Choosing good parameters **manually** is **challenging** and the solution depends strongly on the parameters θ

Question

How to choose suitable parameters?

- Trial & error
- L-curve criterion
- Bilevel Learning (data-driven approach)

Bilevel Learning

Given a training set $\{(\bar{x}_i, y_i)\}_{i=1}^M$ of ground truth and noisy measurements, bilevel learning problem has the following form:

$$\begin{aligned} \min_{\theta \in \mathbb{R}^d} f(\theta) &:= \frac{1}{N} \sum_{i=1}^N g(\hat{x}_i(\theta)) + r(\theta) \\ \text{s.t. } \hat{x}_i(\theta) &:= \arg \min_{x \in \mathbb{R}^N} \Phi_i(x, \theta), \quad i = 1, \dots, N, \end{aligned}$$

where for example:

$$\Phi_i(x, \theta) = \mathcal{D}(Ax, y) + R_\theta(x) \text{ and } g(\hat{x}_i) = \|\hat{x}_i - \bar{x}_i\|^2.$$

Assumptions

1. $\Phi_i(x, \theta)$ is $L_i(\theta)$ smooth and $\mu_i(\theta)$ strongly convex. Moreover, $\nabla_x \Phi_i(x, \theta)$ and $\nabla_x^2 \Phi(x, \theta)$ are continuous in θ .
2. $A(x, \theta) = \nabla_x^2 \Phi(x, \theta)$ is L_A Lipschitz in x and $B(x, \theta) = \nabla_{x\theta}^2 \Phi(x, \theta)$ is L_B Lipschitz in x uniformly for all θ .
3. g is $L_{\nabla g}$ smooth and f is $L_{\nabla f}$ smooth.

Note that the upper-level objective function $f(\theta)$ is nonconvex

Hypergradient

Hypergradient

To simplify, we drop the indices and take $r(\theta) = 0$. Hypergradient is

$$\nabla f(\theta) = \nabla(g \circ \hat{x})(\theta) = \partial \hat{x}(\theta)^T \nabla g(\hat{x}(\theta)).$$

Implicit Function Theorem

If Φ is sufficiently smooth (in x and θ) and strongly convex in x , then $\theta \mapsto \hat{x}(\theta)$ is continuously differentiable with

$$\partial \hat{x}(\theta) := -A(\hat{x}(\theta), \theta)^{-1} B(\hat{x}(\theta), \theta).$$

- We can't calculate $\hat{x}(\theta)$ exactly.
- In large-scale problems (high dimensional x) we solve the linear system $Aq = \nabla g(\hat{x}(\theta))$ approximately (for example using conjugate gradient).

Hypergradient computation

Implicit Function Theorem + CG:

1. Solve lower-level problem to get $\tilde{x}(\theta)$ such that $\|\tilde{x}(\theta) - \hat{x}(\theta)\| \leq \epsilon$
2. Using CG, find q such that $\|A(\tilde{x}(\theta), \theta)q - \nabla g(\tilde{x}(\theta))\| \leq \delta$
3. Calculate the hypergradient estimate $z := -B(\tilde{x}(\theta), \theta)^T q$.

Theorem (Pedregosa (2016); Zucchet & Sacramento (2022))

If ϵ is sufficiently small, then $\|z - \nabla f(\theta)\| = \mathcal{O}(\epsilon + \delta)$.

Theorem (Pedregosa (2016))

Let $\theta \in D \underset{\text{compact}}{\subset} \mathbb{R}^d$. If the sequence $\{\epsilon_k\}_{k=0}^{\infty}$ is positive and $\sum_{k=0}^{\infty} \epsilon_k < \infty$, the sequence of iterates of projected gradient descent $\{\theta_k\}_{k=0}^{\infty}$ given a **fixed** sufficiently small step size has a limit in D and this limit is a stationary point.

How to choose the step size?

Error bound of the Hypergradient

Question

Can we get a computable bound for $e = z - \nabla f(\theta)$?

Ehrhardt & Roberts (2022)

$\|z - \nabla f(\theta)\| = \mathcal{O}(\epsilon + \delta + \epsilon^2 + \delta\epsilon)$. More precisely, at each iteration $k = 1, 2, \dots$, we have

$$\|e_k\| = \|z_k - \nabla f(\theta_k)\| \leq c(\tilde{x}(\theta_k))\epsilon_k + \frac{\|B(\tilde{x}(\theta_k))\|}{\mu}\delta_k + \frac{L_B L_{\nabla g}}{\mu}\epsilon_k^2,$$

where $c(\tilde{x}(\theta_k))$ has a computable closed form.

Hyperparameter Optimisation Algorithms

Descent direction & backtracking line search

suppose we have the descent update

$$\theta_{k+1} = \theta_k - \alpha_k z_k, \quad k = 1, 2, \dots$$

Question

When is $-z_k$ a descent direction?

Proposition

If $\frac{\|e_k\|}{\|z_k\|} < 1$ and $\|z_k\| \neq 0$, then $-z_k$ is a descent direction.

Armijo backtracking line search: At each iteration k , for a given direction z_k and a starting step size β find the smallest $i_k \in \mathbb{N}$ such that $f(\theta_k - \beta \rho^{i_k} z_k) \leq f(\theta_k) - \eta \beta \rho^{i_k} \nabla f(\theta_k)^T z_k$, for $0 < \rho, \eta < 1$. After finding i_k , we denote the step size by $\alpha_k := \beta \rho^{i_k}$.

We don't have access to $f(\theta)$ and $\nabla f(\theta)$!

Inexact backtracking line search

Lemma 1 (New)

Let $\frac{\|e_k\|}{\|z_k\|} \leq 1 - \eta$, g be convex, and aforementioned assumptions are satisfied. If

$$g(\tilde{x}(\theta_{k+1})) + \|\nabla g(\tilde{x}(\theta_{k+1}))\| \epsilon_{k+1} + L_{\nabla g} \epsilon_{k+1}^2 - g(\tilde{x}(\theta_k)) + \|\nabla g(\tilde{x}(\theta_k))\| \epsilon_k \leq -\eta(2 - \eta)\alpha_k \|z_k\|^2,$$

where $\alpha_k = \beta_k \rho^{i_k}$ and $\theta_{k+1} = \theta_k - \alpha_k z_k$ for $\beta_k > 0$, $i_k \geq 0$, and $0 < \rho, \eta < 1$, then the Armijo backtracking line search for f holds and sufficient decrease in the exact function evaluation occurs.

In this line search rule all of the components are computable.

Lemma 2 (New)

Assuming $\|z_k\| > 0$, $0 < \eta < 1$, and $\frac{\|e_k\|}{\|z_k\|} \leq 1 - \eta$, denoting $h(a, b) := a + \sqrt{a^2 + b}$, for all

$$\epsilon_k \in \left[0, h \left(\frac{-\|\nabla g(\tilde{x}(\theta_k))\|}{L_{\nabla g}}, \frac{(\eta - \eta^2)^2 \|z_k\|^2}{4L_{\nabla g}L_{\nabla f}} \right) \right],$$

there exist $0 \leq \alpha_{\min}(\epsilon_k) < \alpha_{\max}(\epsilon_k)$ such that for all $\alpha_k \in [\alpha_{\min}(\epsilon_k), \alpha_{\max}(\epsilon_k)]$ the sufficient decrease condition (Lemma 1) holds.

Corollary

If $\|\nabla f(\theta_k)\| > 0$, there exist a range of ϵ_k independent from z_k and $\tilde{x}(\theta_k)$ such that the inclusion

$$\epsilon_k \in \left[0, h \left(\frac{-\|\nabla g(\tilde{x}(\theta_k))\|}{L_{\nabla g}}, \frac{(\eta - \eta^2)^2 \|z_k\|^2}{4L_{\nabla g} L_{\nabla f}} \right) \right)$$

and consequently Lemma 2 hold.

Does the introduced backtracking line search find a step size $\alpha_k \in [\alpha_{\min}(\epsilon_k), \alpha_{\max}(\epsilon_k)]$?

Lemma 3: (Existence of step size)

For each iteration $k \geq 1$ of $\theta_{k+1} = \theta_k - \alpha_k z_k$ using our line search rule (Lemma 1), given fixed $0 < \beta$ and $0 < \rho < 1$, for all $\epsilon_k \geq 0$ sufficiently small, there exists an $n \in \mathbb{N}$ such that $\alpha_{\min}(\epsilon_k) \leq \beta \rho^n \leq \alpha_{\max}(\epsilon_k)$.

Theorem (New)

Under the aforementioned assumptions, if at each iteration we pick α_k and ϵ_k such that $\alpha_k > \rho\alpha_{\max}(\epsilon_k)$ and $\alpha_k \in [\alpha_{\min}(\epsilon_k), \alpha_{\max}(\epsilon_k)]$, and a direction z_k for which error bound holds and $\|z_k\| \neq 0$, assuming that $\frac{\|e_k\|}{\|z_k\|} \leq 1 - \eta$ and f is bounded below, we have

$$\lim_{k \rightarrow \infty} \|z_k\| = 0,$$

and

$$\lim_{k \rightarrow \infty} \|\nabla f(\theta_k)\| = 0.$$

Dynamic hyperparameter optimization with inexact line search (DHOILS)

While the stopping criterion is not satisfied:

1. If $\epsilon > \bar{\epsilon}$:
 $\epsilon \leftarrow \bar{\epsilon}$ and recalculate the Inexact
 hypergradient
2. Perform Inexact Backtracking Line Search.

Inexact Backtracking Line Search

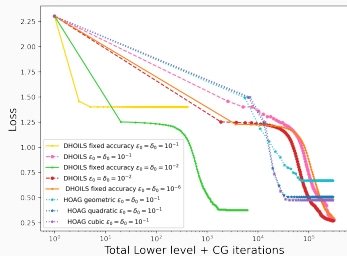
While the maximum number of backtracking is
not reached:

1. If a step size $\alpha = \rho^i \beta$ satisfies the introduced
inexact line search rule:
 return $\theta_{\text{new}} = \theta - \alpha z$
2. Decrease ϵ and recalculate the hyper gradient.
3. Restart the backtracking loop with new z, ϵ, δ ,
and $\max_{\text{iter}} + 1$.

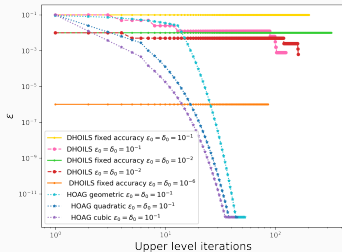
Numerical results

Multinomial logistic regression problem

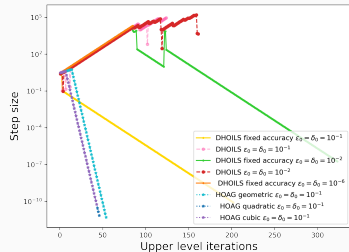
$$\min_{\theta \in \mathbb{R}^d} f(\theta) := \sum_{i=1}^N \psi(\bar{b}_i, \bar{a}_i^T \hat{x}(\theta)) \quad \text{s.t.} \quad \hat{x}(\theta) := \arg \min_{x \in \mathbb{R}^{p \times q}} \sum_{j=1}^M \psi(b_j, a_j^T x) + \frac{1}{2} \sum_{k=1}^d e^{\theta_k} x_k^2$$



(a) Loss vs total lower-level computational cost



(b) Accuracy of the lower-level solver at each upper-level iteration



(c) Upper-level step size

2D TV denoising

$$\min_{\theta} \frac{1}{2} \|\hat{x}(\theta) - \bar{x}\|^2 \quad \text{s.t.} \quad \hat{x}(\theta) = \arg \min_x \frac{1}{2} \|x - y\|^2 + e^{\theta_1} \sum_{i,j} \sqrt{|\nabla x_i|^2 + |\nabla x_j|^2 + (e^{\theta_2})^2} + \frac{e^{\theta_3}}{2} \|x\|^2$$



(a) Ground truth



(b) Noisy image $\sigma = 0.1$
PSNR = 19.99

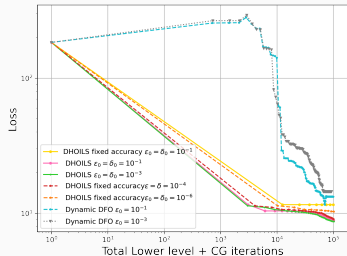


(c) DHOILS
PSNR = 29.24

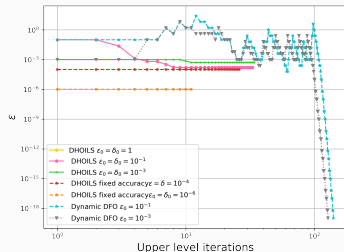


(d) DFO-LS
PSNR = 27.3

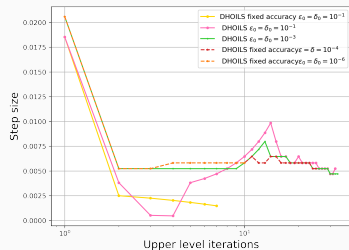
2D TV denoising



(a) Loss vs total lower-level computational cost



(b) Accuracy of the lower-level solver at each upper-level iteration



(c) Upper-level step size

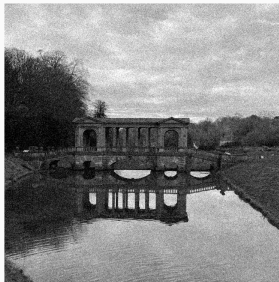
Field of Experts denoising

$$\min_{\theta} \frac{1}{2} \|\hat{x}(\theta) - \bar{x}\|^2 \quad \text{s.t.} \quad \hat{x}(\theta) = \arg \min_x \left\{ \frac{1}{2} \|x - y\|^2 + e^{\theta_0} \sum_{k=1}^K e^{\theta_k} \|c_k * x\|_{\theta_{K+k}} + \frac{\mu}{2} \|x\|^2 \right\},$$

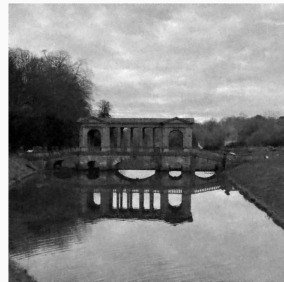
We used 30 filters of size 7×7 .



(a) Ground truth

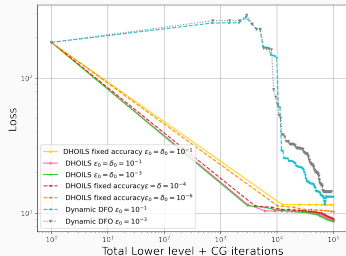


(b) Noisy image $\sigma = 0.1$
PSNR = 20.01

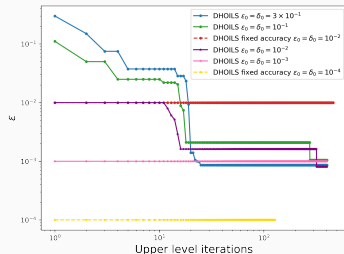


(c) DHOILS
PSNR = 28.68

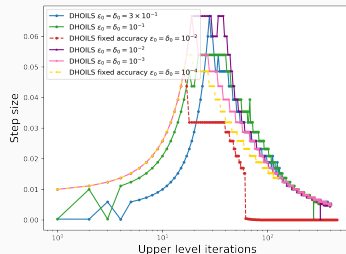
Field of Experts denoising



(a) Loss vs total lower-level computational cost



(b) Accuracy of the lower-level solver at each upper-level iteration



(c) Upper-level step size

Conclusions

- Utilizing the error bound to find a descent direction under inexactness.
- Proposing an inexact backtracking line search with approximate function evaluations and an approximate hypergradient.
- Developing a dynamic algorithm integrated with a provably convergent line search, which determines the required accuracy and a suitable stepsize.
- Outperforming HOAG and dynamic DFO-LS while maintaining robustness against the choice of initial accuracy and starting step size.

Future work

- Performing a worst-case complexity analysis
- Exploring randomisation in this context (stochastic case)

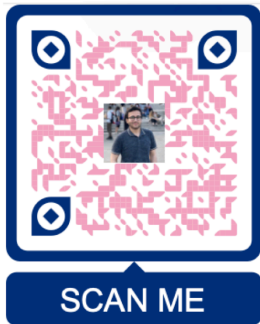
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